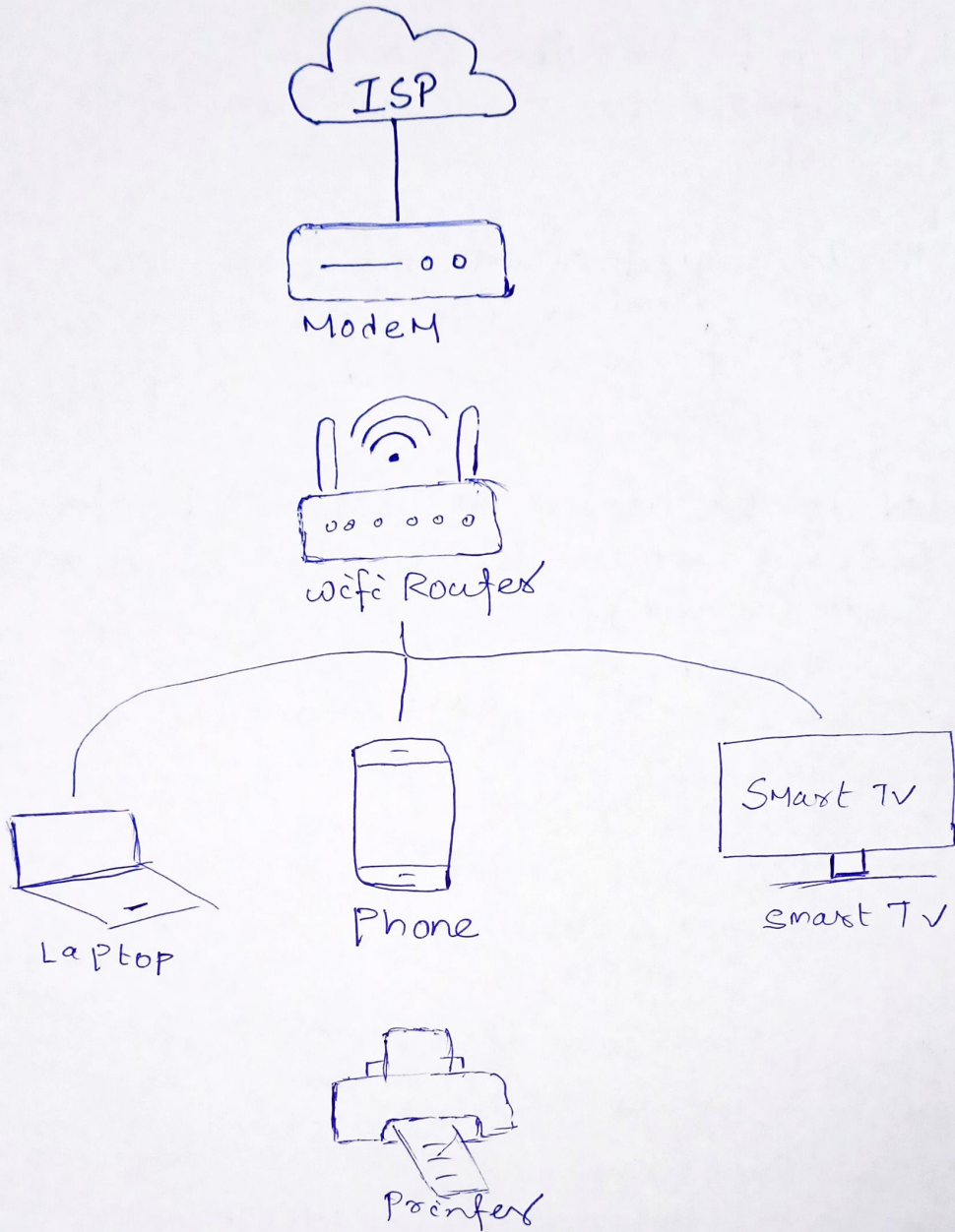


Assignment - 01



Explanation

- **ISP (Internet Service Provider):**
Provides internet access to your home. Example: Jio, Airtel, BSNL.
- **Modem:**
Converts the signal from the ISP into a digital signal your home devices can use. It connects directly to the ISP through a wired line (fiber optic/cable/DSL).
- **Wifi Router**
connected to the modem. It distributes internet to multiple devices either through wifi (wireless) or LAN cables (wired).
- **Devices connected:**
 - Laptop (via Wi-Fi or LAN cable)
 - Mobile phones (Wi-Fi)
 - Smart TV (Wi-Fi)
 - Printer (Can be connected to the router wirelessly or wired to the laptop).

Important Points:

- Router assigns private IP address to each device inside the house using DHCP.
- All devices inside the home share one public IP address on the Internet (using NAT).
- Wireless devices (like phones) and wired device (like a laptop via LAN) co-existably.